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The formation of the DNA double helix creates opposing changes in entropy and enthalpy. Favorable bonding interactions via hydrogen bonds during Watson-Crick base pairing results in negative enthalpy, and restricted rotation and flexibility of the DNA backbone generates negative entropy. Scientists hypothesize that hydrogen bonding between bases not only stabilizes the double helix but is also crucial for selective and efficient replication.

Analogs that are similar in size and shape to naturally occurring bases can be used to determine the influence of hydrogen bonding on base pair selectivity. To mimic the structure of deoxythymidine triphosphate (dTTP), researchers synthesized dNTP derivatives of difluorotoluene (dFTP). dFTP is a nonpolar analog of dTTP that lacks Watson-Crick hydrogen bonding. Klenow fragment polymerase (KF), which has 3'-5' but not 5'-3' exonuclease activity, was incubated with a mixture of DNA template, primers, and dNTPs, including dFTP. The efficiency of dFTP and natural dTTP nucleotide incorporation into a growing primer strand by KF is shown in Figure 1.

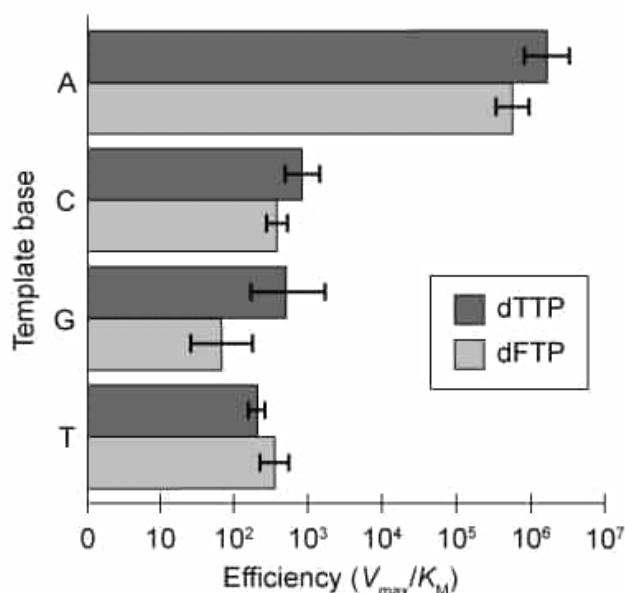


Figure 1 Template-specific selection of dFTP and dTTP by the KF enzyme

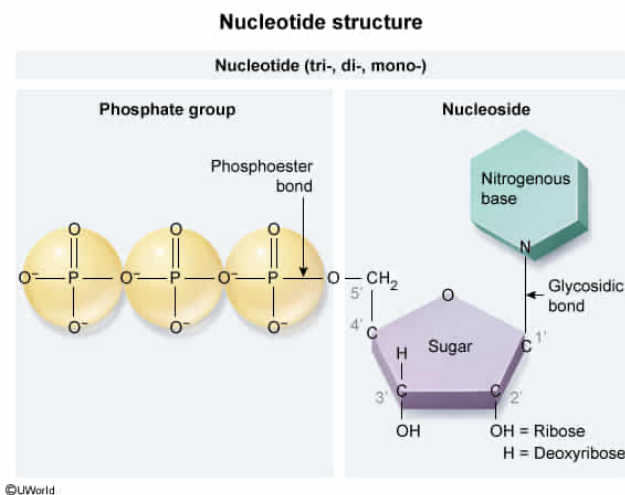
Which of the following would NOT be present in the structure of the dFTP nucleotide used for replication?

- ☐ A. Bond to phosphate on the 5' carbon of the pentose sugar [6%]
- ☐ B. Glycosidic bond on the 1' carbon of the pentose sugar [10%]
- ✓ ☒ C. OH group on the 2' carbon of the pentose sugar [71%]
- ☐ D. H group on the 5' carbon of the pentose sugar [11%]

Correct

71% answered correctly

Explanation:



Deoxyribonucleotides are the monomer units that make up the structure of DNA. A **nucleotide** is composed of a nucleoside molecule attached by a **phosphoester bond** to one or more phosphate groups. **Nucleosides** consist of a **pentose** (five-carbon) sugar linked to a **nitrogenous base** on the 1' carbon by a covalent glycosidic bond.

Ribose and deoxyribose are the two types of pentose sugars used in nucleotide production, differing only in structure at the **2' carbon**. Ribose sugars have a hydroxyl group at the 2' carbon where deoxyribose sugars have a **hydrogen atom**. DNA replication only utilizes nucleotides that contain **deoxyribose sugars**. The dFTP nucleotide is a deoxyribonucleotide (dNTP) where the nitrogenous base is replaced with difluorotoluene; all other structural components remain unaltered. Because the remaining nucleotide structure is unaltered, dFTP nucleotides would *not* contain hydroxyl groups on the 2' carbon of the deoxyribose pentose sugar.

(Choice A) All nucleotides, including those made with difluorotoluene, contain phosphate groups on the 5' carbon of the pentose sugar.

(Choice B) All nucleotides and nucleosides are bonded to nitrogenous bases at the 1' carbon of the pentose sugar through a glycosidic bond, which is a bond between the anomeric carbon of a carbohydrate and any other group.

(Choice D) The H group on the 5' carbon of the pentose sugar is present in nucleosides and nucleotides whether they are made with ribose or deoxyribose.

Educational objective:

A nucleoside is a pentose (five-carbon) sugar linked to a nitrogenous base on the 1' carbon by a covalent glycosidic bond. Nucleotides consist of a nucleoside attached to one or more phosphate groups by a phosphoester bond. If the pentose has a hydroxyl group at the 2' carbon, it is ribose; if only a hydrogen is present at the same position, it is deoxyribose.

Time spent:0 sec

Subject:

Biochemistry

QID:402149

System:

Carbohydrates, Nucleotides, and Lipids

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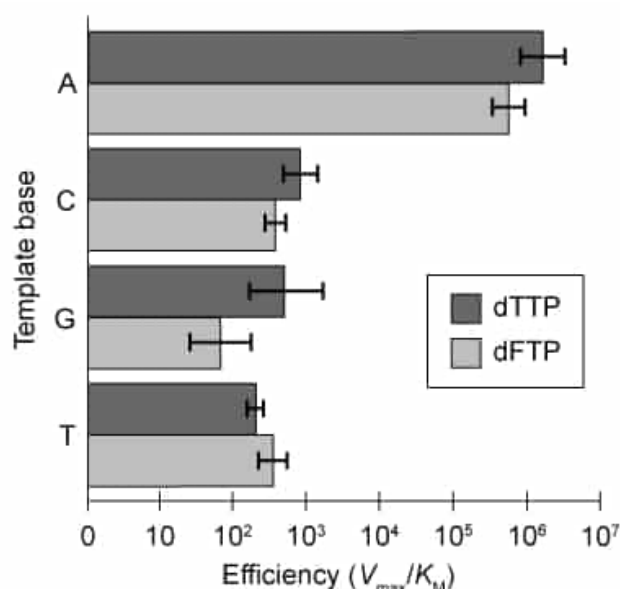


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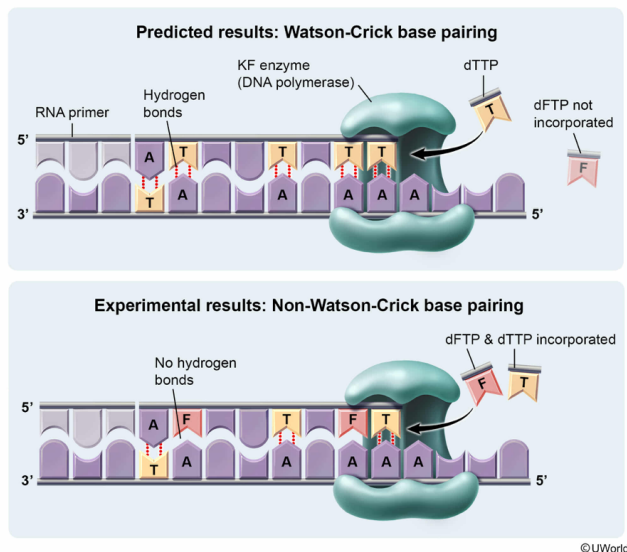
The experimental results described in the passage are inconsistent with which hypothesis?

- ☐ A. dFTP can serve as a substrate for DNA polymerase. [7%]
- ✓ ☒ B. Watson-Crick hydrogen bonds are required for base pair selectivity. [70%]
- ☐ C. dFTP will serve as an effective template for DNA replication. [14%]
- ☐ D. Molecular shape is involved in base pair selectivity. [6%]

Correct

71% answered correctly

Explanation:



In DNA replication, dNTPs containing thymine (dTTP) are **selectively** inserted opposite adenine nucleotides rather than guanine, cytosine, or other thymine nucleotides in the template strand. Scientists generally attribute this selectivity to the **hydrogen bonding** between thymine and adenine in the Watson-Crick model of DNA.

The study described in the passage tested this hypothesis by synthesizing dFTP, a dNTP analog where the nitrogenous base is replaced with difluorotoluene, which lacks Watson-Crick hydrogen bonding capacity but is similar in size and shape to thymine. The effect of hydrogen bonding on base selectivity was measured as the efficiency with which dTTP and dFTP were inserted into a growing primer strand by KF.

Scientists predicted that only dTTP nucleotides would be selectively paired with adenine, whereas dFTP would not be added to the growing strand. However, the

results depicted in Figure 1 show no significant difference in the base selectivity between dTTP and its structural analog dFTP. When KF detected adenine in the template strand, dFTP was inserted with the same efficiency as dTTP. Similarly, when the template base was guanine, cytosine, or thymine, the efficiency of dFTP and dTTP insertion was greatly reduced. Therefore, dFTP exhibited the same base pair selectivity as dTTP despite its inability to form Watson-Crick hydrogen bonds.

(Choice A) dFTP was efficiently inserted into DNA strands opposite adenine; therefore, it serves as a good substrate for DNA polymerase.

(Choice C) The template strand in the experiment did not contain dFTP nucleotides, so the results are neither consistent nor inconsistent with the hypothesis that dFTP is an effective template nucleotide.

(Choice D) dFTP is similar in shape to dTTP. Its selective incorporation into DNA suggests that shape is involved in selectivity.

Educational objective:

DNA polymerase selectively inserts complementary nucleotides during DNA replication. Complementary nucleotides form hydrogen bonds with bases on the opposite strand. The Watson-Crick model shows that adenine is paired with thymine whereas guanine is paired with cytosine.

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Biochemistry

QID:402150

System:
Carbohydrates, Nucleotides, and Lipids

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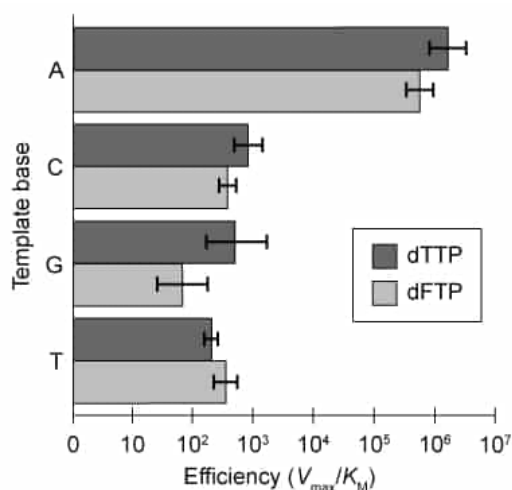


Figure 1 Template-specific selection of dFTP and dTTP by the KF enzyme

In the DNA polymerase active site, which nucleotide base would provide the most Watson-Crick donors for interactions with amino acid residues?

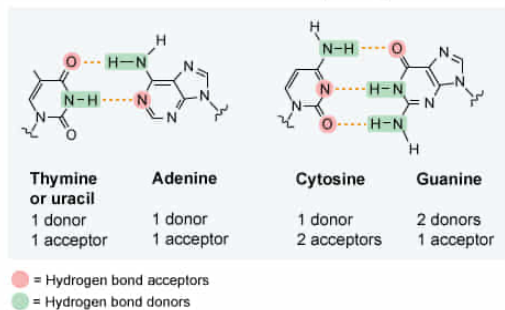
- ☐ A. Adenine [27%]
- ☐ B. Cytosine [12%]
- ☒ C. Guanine [53%]
- ☐ D. Thymine [6%]

Correct

53% answered correctly

Explanation:

Watson-Crick base pairing



Hydrogen bonds form between **donor hydrogen** atoms and **acceptor oxygen and nitrogen** atoms. Donor hydrogen atoms must be covalently bonded to **electronegative atoms** such as nitrogen or oxygen, which confer a strong partial positive charge to the hydrogens, and acceptor O or N atoms must have at least **one lone pair of electrons**. In **Watson-Crick** base pairing, hydrogen bonds form between particular acceptor atoms of one base and donor atoms of the complementary base, and vice versa.

Adenine (A) and thymine (T), which each have **1 donor and 1 acceptor atom**, form two hydrogen bonds between them (note that thymine has two potential acceptors in the form of carbonyls, but only one participates in Watson-Crick pairing). **Guanine (G) and cytosine (C)** pairs form three hydrogen bonds: Guanine contributes **1 acceptor and 2 donors** whereas cytosine contributes **2 acceptors and 1 donor**.

The passage states that amino acids in the active site of DNA polymerase can interact with nucleotides by forming hydrogen bonds with Watson-Crick donor and acceptor atoms. Guanine has two Watson-Crick donors; therefore, it contributes more donors for bonding to active site amino acids than any other nucleotide base.

(Choices A, B, and D) Adenine, cytosine, and thymine each have only one Watson-Crick donor and would contribute fewer donors for bonding to active site residues than guanine.

Educational objective:

Hydrogen bond acceptors are electronegative atoms (nitrogen or oxygen) that have at least one lone pair of electrons, and hydrogen bond donors are hydrogen atoms bound to electronegative atoms. Watson-Crick base pairing occurs when particular hydrogen bond donors and acceptors in one nucleotide bond to acceptors and donors in the complementary nucleotide. Adenine and thymine each have 1 Watson-Crick donor and 1 acceptor, whereas guanine has 2 donors and 1 acceptor atom, and cytosine has 1 donor and 2 acceptors.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402151

System:
Carbohydrates, Nucleotides, and Lipids

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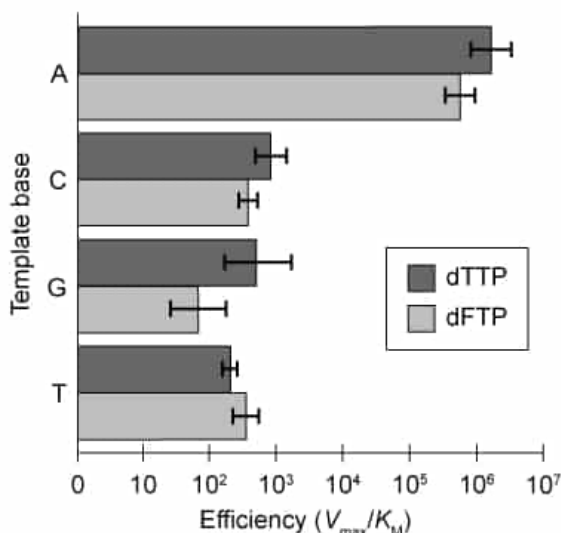


Figure 1 Template-specific selection of dFTP and dTTP by the KF enzyme

The Klenow fragment used in the experiment would be able to perform which of the following repair processes?

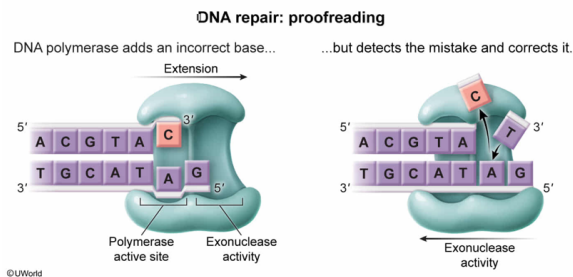
- ☐ A. Correction of mismatched nucleotides in the middle of a completed strand [6%]
- ☒ B. Replacement of nucleotides at the 3' end of the growing strand [49%]

- ☐ C. Excision of thymine dimers at the 5' end of the growing strand [35%]
- ☐ D. Removal of damaged bases from the middle of the template strand [8%]

Correct

50% answered correctly

Explanation:



DNA polymerases extend growing strands of DNA to form double-stranded, reverse complementary DNA. If an error is made during this process, many polymerases are capable of detecting and removing incorrect nucleotides at the ends of a growing DNA strand. This function, referred to as **proofreading**, prevents mutations or errors in DNA replication from accumulating.

Normally, DNA polymerases are equipped with both 5'-3' and 3'-5' **exonuclease activity** that allows them to remove and replace incorrect nucleotides at either **end of a DNA strand**. However, the passage states that the Klenow fragment (KF) enzyme described in the experiment does not have 5'-3' exonuclease activity. KF can only proofread DNA in the **3'-5' direction** on the growing strand, so only errors at the **3' end of the growing strand** can be repaired.

(Choice A) **Mismatch repair enzymes**, which have *endonuclease* rather than *exonuclease* activity, are required to correct mismatched nucleotides within a DNA strand. Endonucleases can cleave phosphodiester bonds in the middle of nucleic acid strand.

(Choice C) KF would not remove thymine dimers from the 5' end of section 1 because it is deficient in 5'-3' exonuclease activity. **Nucleotide repair enzymes** are required to excise thymine dimers.

(Choice D) Damaged bases are removed after replication by **base excision repair** enzymes with *endonuclease* activity.

Educational objective:

DNA polymerase I proofreads DNA and normally has exonuclease activity in the 5'-3' as well as the 3'-5' direction that allows it to remove primers and damaged or incorrect bases at the ends of the strand. It cannot fix mistakes in the middle of a strand; instead, base excision repair and nucleotide excision enzymes have endonuclease activity to remove damaged bases and mismatched nucleotides from the middle of a DNA strand, respectively.

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Subject:

Biochemistry

QID:402152

System:

Carbohydrates, Nucleotides, and Lipids

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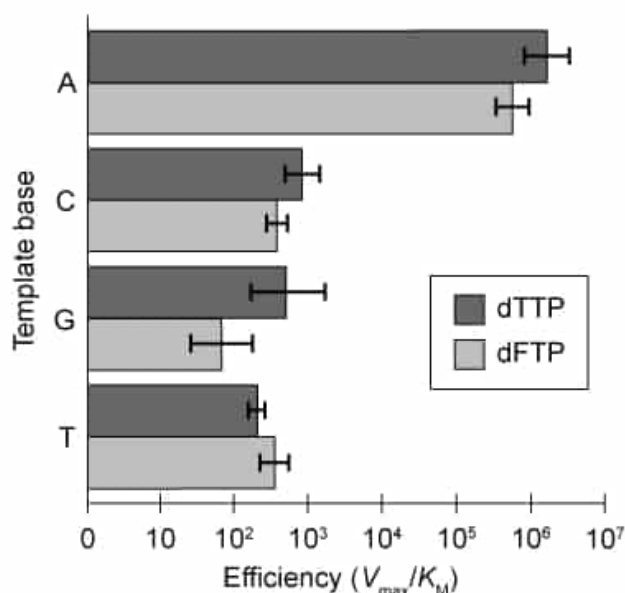


Figure 1 Template-specific selection of dFTP and dTTP by the KF enzyme

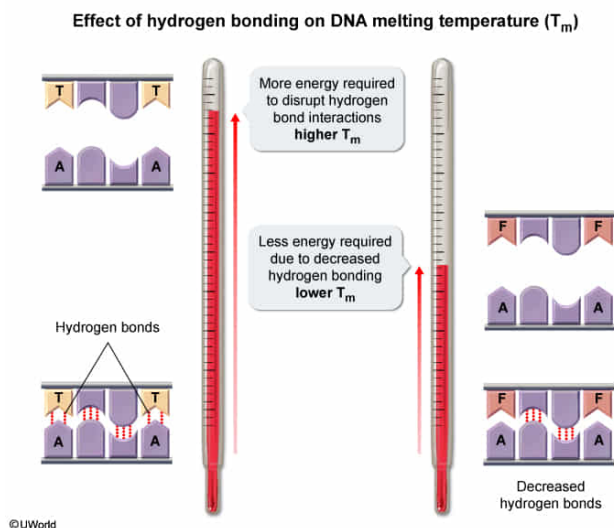
Based on the information in the passage, what effect would incorporation of dFTP in place of dTTP most likely have on the melting temperature of a DNA molecule of a fixed length?

- ☐ A. The melting temperature would increase due to reduced base stacking. [1%]
- ✓ ☒ B. The melting temperature would decrease due to decreased hydrogen bonding. [88%]
- ☐ C. The melting temperature would not change because dFTP is similar to dTTP. [8%]
- ☐ D. The effect on melting temperature depends on the number of nucleotides in the DNA strand. [1%]

Correct

88% answered correctly

Explanation:



The **melting temperature** T_m of DNA is the temperature at which 50% of the double helices in solution have been **separated into single strands**. It is determined by the strength of the **intermolecular forces** holding the strands together. An increased number of intermolecular interactions increases the T_m because more energy is required to disrupt them.

Hydrogen bonding between bases (**Watson-Crick hydrogen bonding**) is one of the major intermolecular forces that hold DNA strands together. The number of hydrogen bonds between base pairs differs: G and C share three hydrogen bonds whereas A and T share only two. Therefore, DNA molecules with high GC content have higher melting temperatures than those with low GC content.

The passage states that dFTP lacks Watson-Crick hydrogen bonding. Although the enzyme was able to incorporate dFTP based on the shape of the nitrogenous base and the enzyme active site, *stability* of the double helix depends on hydrogen bonds between the template strand and the bases of the complementary strand. Therefore, dFTP incorporation in place of dTTP will weaken interactions with adenine. Because there are **fewer hydrogen bonds** holding the strands together, less energy will be required to separate them. Consequently, DNA that incorporated dFTP instead of dTTP will most likely have **a decreased melting temperature**.

(Choice A) The melting temperature is expected to *decrease*. Reduced base stacking would most likely *decrease*, not increase, T_m due to decreased hydrophobic interactions.

(Choice C) Although dFTP is similar in shape to dTTP and specifically pairs with adenine, its inability to form hydrogen bonds with adenine makes the interaction weaker and results in a decreased melting temperature.

(Choice D) The melting temperature of a DNA molecule is affected by the number of nucleotides. DNA molecules with more nucleotides have more intermolecular interactions (eg, hydrogen bonds) and increased melting points. However, the question states that the DNA molecule has a fixed length, so the number of nucleotides is not a factor in this scenario.

Educational objective:

The strands of a DNA double helix are held together by intermolecular interactions that include hydrogen bonds. Reducing the number of hydrogen bonds decreases the melting point of a double helix.

Time spent:0 sec

Subject:
Biochemistry

QID:402153

System:
Carbohydrates, Nucleotides, and Lipids

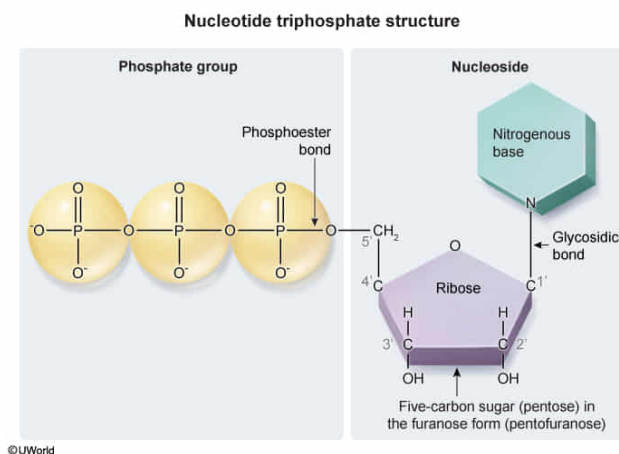
Which of the following most accurately describes the ribose component of a nucleotide triphosphate?

- ✔ ☒ A. It is a pentofuranose. [63%]
☐ B. It is a pentopyranose. [21%]
☐ C. It is a hexofuranose. [8%]
☐ D. It is a hexopyranose. [6%]

Correct

62% answered correctly

Explanation:



Nucleotide triphosphates (NTPs) consist of a **nitrogenous base** (adenine, guanine, cytosine, or uracil), a **sugar** (ribose), and a **triphosphate**. **Ribose is a pentose** (five-carbon sugar) that links to the nitrogenous base through a **glycosidic bond**. A **phosphoester bond** links the 5' carbon of ribose to the triphosphate group.

Carbohydrates can participate in glycosidic linkages only when in cyclic form (**furanose or pyranose**), so ribose must be cyclic in NTPs. In the furanose form, the 5' hydroxyl group can readily participate in phosphoester bonds. Because the ribose in an NTP is a **pentose in the furanose form**, it is classified as a **pentofuranose**.

(Choice B) In the pyranose form, the 5' hydroxyl group of ribose becomes part of the ring structure and **cannot participate** in phosphoester bonds, so ribose cannot be in the pyranose form in NTPs.

(Choices C and D) Ribose is a pentose (five carbons), not a hexose (six carbons).

Educational objective:

Ribonucleotide triphosphates contain a nitrogenous base, ribose, and a triphosphate group linked to the 5' carbon of ribose. Ribose is a five-carbon sugar (pentose) that must adopt the furanose form to be incorporated into the nucleotide triphosphates. In this form, it can be classified as a pentofuranose.

Time spent:0 sec**Subject:**
Biochemistry**QID:**402154**System:**
Carbohydrates, Nucleotides, and Lipids